

5/31/2018

Print

SG-391-500

Engine Test Certificate

Perkins Engine Company Limited
Eastfield, Peterborough PE1 5FQ
Great Britain

May 31, 2018

 Perkins

This engine has been tested to its maximum rating in accordance with Perkins defined requirements and has performed satisfactorily.

The electronics on this engine have been preset and when the engine has been run-in, the gross power output will be within $\pm 3\%$ of the gross rating listed below, provided the exhaust and induction systems, fuel and lubricating oils are in accordance with Perkins defined requirements.

Engine Type	Perkins Serial Number	Customer
2506A-E15TAG2	MGBF5083N09694D	LPS2

Reference	Technical Data
Emission Capability	UNREGULATED Low SFC
Engine Speed	50 Hz @1500RPM
Engine Type	2506A-E15TAG2
Fuel Consumption - Prime Power	192 g/kWH
Fuel Consumption - Standby Max	193 g/kWH
Gross Engine Power - Prime Power	458 kW
Gross Engine Power - Standby Max	503 kW
Gross Rating	458 kW
Net Rating	435 kW

This engine was built and tested in accordance with our approved ISO 9001 Quality Program and stated Rating, as determined by the ISO 3046 (BS5541) Standard.

Performance certified as satisfactory.

Signed,

For and on behalf of Perkins Engines Company Limited - Large Engines Division.

MUST BE KEPT WITH ENGINE AT ALL TIMES

STAMP

Test Certificate

FOR

SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR

Machine No. **N19F230772**

Type Test has been conducted on similar design machine
STANDARDS: Generator generally conforms to BS 5000 : Part 3,
 Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
 Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: HCI544D1	AVR: AS440	AVR NO: 1155560679	ENCLOSURE: IP23
RATED KVA: 500	RATED VOLTS: 415	RATED AMPS: 695.6	FREQUENCY: 50
POWER FACTOR: .8	STR CONN: Series Star	STR WINDING: 311	DUTY: Base Continuous
EXC. VOLTS: 48	EXC.AMPS: 2.3	AMB TEMP: 40	

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION
STATOR	0.01212	2+	OK	
ROTOR	1.904	2+	OK	

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP
52	416.4	0.00	0.00	0.00	0.00	10	.55
50	415	695.6	500	.8	400	48	2.3

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.HC5.4.D.1.311.,0.5.,14.,AS440.Black - RAL9011
 Remarks:

Date: 06-JUN-19

Job No: 3721733

AUTHORISED BY

Shubham S Pawar
 Quality Assurance

Certificate Of Conformity

Quality System Certification
 ISO 9001:2015

Frame Size : **HCI544D1**Serial Number : **N19F230772**KVA: **500**

Manufactured Date : 06-JUN-19

EU DECLARATION OF CONFORMITY



**Generator
Technologies**

This synchronous A.C. generator is designed for incorporation into an electricity generating-set and fulfils all the provisions of the following EU Directive(s) when installed in accordance with the installation instructions contained in the product documentation:

2014/35/EU	Low Voltage Directive
2014/30/EU	The Electromagnetic Compatibility (EMC) Directive
2011/65/EU	Restriction on Hazardous Substances in Electrical and Electronic Equipment (RoHS) Directive
2015/863	Delegated Directive amending Annex II of 2011/65/EU

and that the standards and/or technical specifications referenced below have been applied:

EN 61000-6-2:2005	Electromagnetic compatibility (EMC). Generic standards - Part 6-2: Immunity for industrial environments
EN 61000-6-4:2007+A1:2011	Electromagnetic compatibility (EMC). Generic standards - Part 6-4: Emission standard for industrial environments
EN ISO 12100:2010	Safety of machinery - General principles for design - Risk assessment and risk reduction
EN 60034-1:2010	Rotating electrical machines - Part 1: Rating and performance
BS ISO 8528-3:2005	Reciprocating internal combustion engine driven alternating current generating sets - Part 3: Alternating current generators for generating sets
BS 5000-3:2006	Rotating electrical machines of particular types or for particular applications - Part 3: Generators to be driven by reciprocating internal combustion engines - Requirements for resistance to vibration
EN 50581:2012	Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances

This declaration has been issued under the sole responsibility of the manufacturer. The object of this Declaration is in conformity with the relevant Union harmonization Legislation.

The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Generator Technologies Limited, Fountain Court, Lynch Wood, Peterborough, PE2 6FZ, UK.

Signed:

Date: 14th September 2018

Name, Title and Address:

Kevan J Simon
Global Technical Director
Cummins Generator Technologies
Fountain Court
Lynch Wood
Peterborough,
PE2 6FZ, UK.

EU DECLARATION OF CONFORMITY



**Generator
Technologies**

The A.C. Generator utilizes hazardous material exemptions as detailed in Annex III of EU Directive 2011/65/EU

Products carrying the following descriptions are considered to be out of scope of RoHS Directive 2011/65/EU, intended to be installed in Large Scale Fixed Installations and for installation into a predefined and dedicated location, installed and de-installed by professionals:

LVI80*

LVSI80*

DSG 99*

DSG 114*

DSG 125*

DSG 144*

Where "*" represents any combination of letters and characters completing the specific description of the product